YETI PAW v1.3

Lateral raise handle — assembly sheets

Designed by **Schnibbz** · YETI STRENGTH

1 · Overview (assembled)

2 · Handle assembly

3 · Yoke assembly

3b · Yoke pivot — section

4 · Pad assembly

v1.3: two parts changed. The drawings in sheets 2 and 3 still show the v1.2 parts.
Handle bearing: use the plain **R8-2RS** (no flange), not the flanged FR8 drawn in sheet 2. Press it in from the outer face until it stops on the ledge behind the seat; its face ends flush with the plate.
Carabiner eye sleeve: use the solid bronze plain sleeve, **McMaster 6391K256** (3/4" bore × 7/8" OD × 1/2" long), not the flanged sleeve drawn in sheet 3. Press it through the eye until it sits flush with both faces.

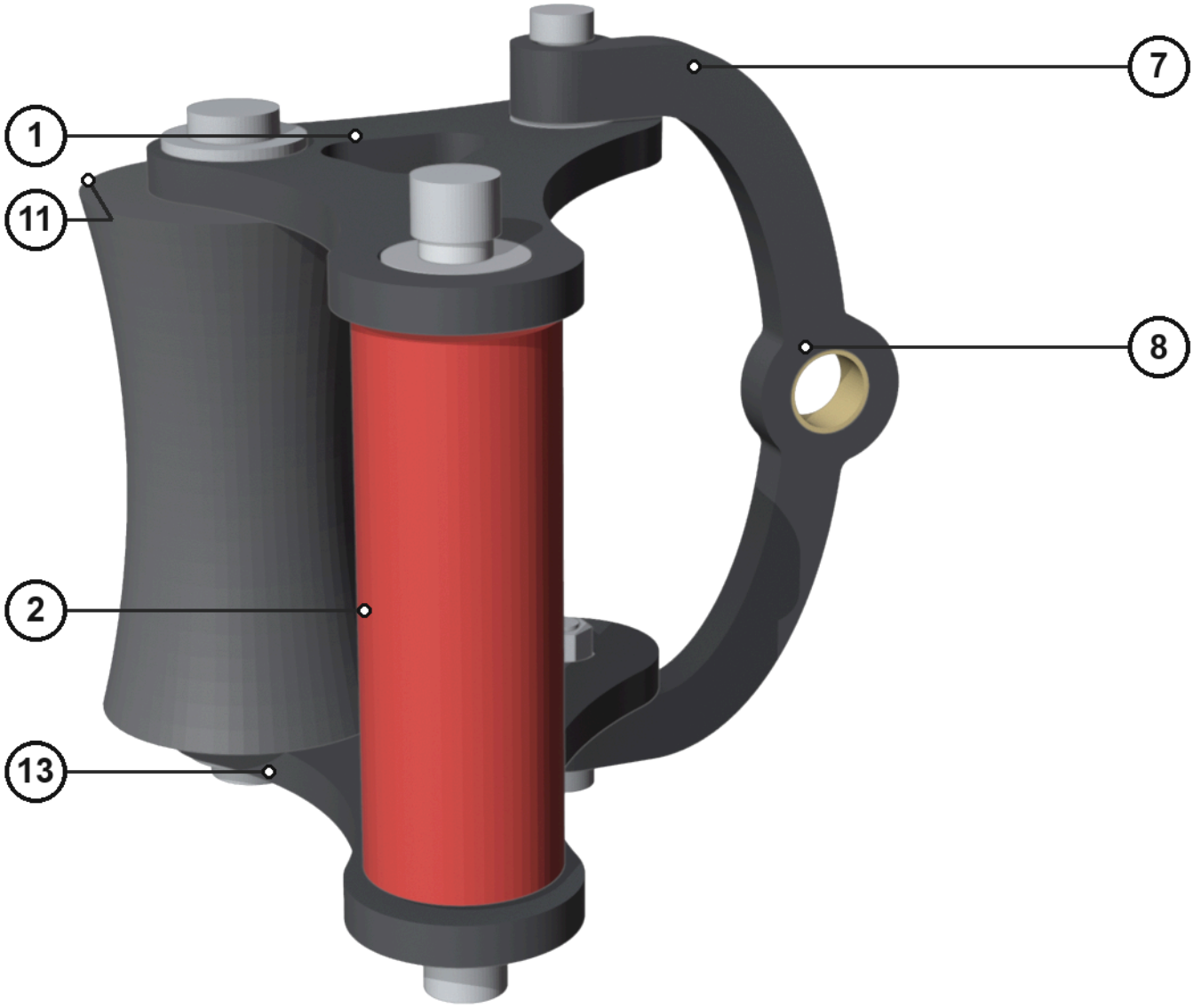
Before you start: print the parts with the settings in PRINT-SETTINGS.md. Clean the yoke's two Ø12 pivot bores back to round with a Ø12 mm drill bit turned by hand. Run a 1/2"-13 bolt through both dowel threads. Bearings and sleeves are press fits: **no Loctite or CA glue on plastic** (it attacks PETG); epoxy is OK if a bearing seat is loose. Blue Loctite goes on the handle screws only, which are metal into metal. Assembly order: see the README.

Not professionally engineered. The Yeti Paw was designed by a hobbyist, not by a licensed or qualified engineer. Any strength or safety-factor figures are amateur estimates from a simplified computer model, not a certified load rating, and come with no guarantee.

Build and use at your own risk. You are responsible for printing, assembling, inspecting and using this device. Print quality, materials, settings and assembly vary, so your parts may be weaker than described. Inspect every part before each use and stop using it at the first sign of cracking, play or damage. The designer accepts no liability for injury or damage arising from building or using it.

1 · Overview (assembled)

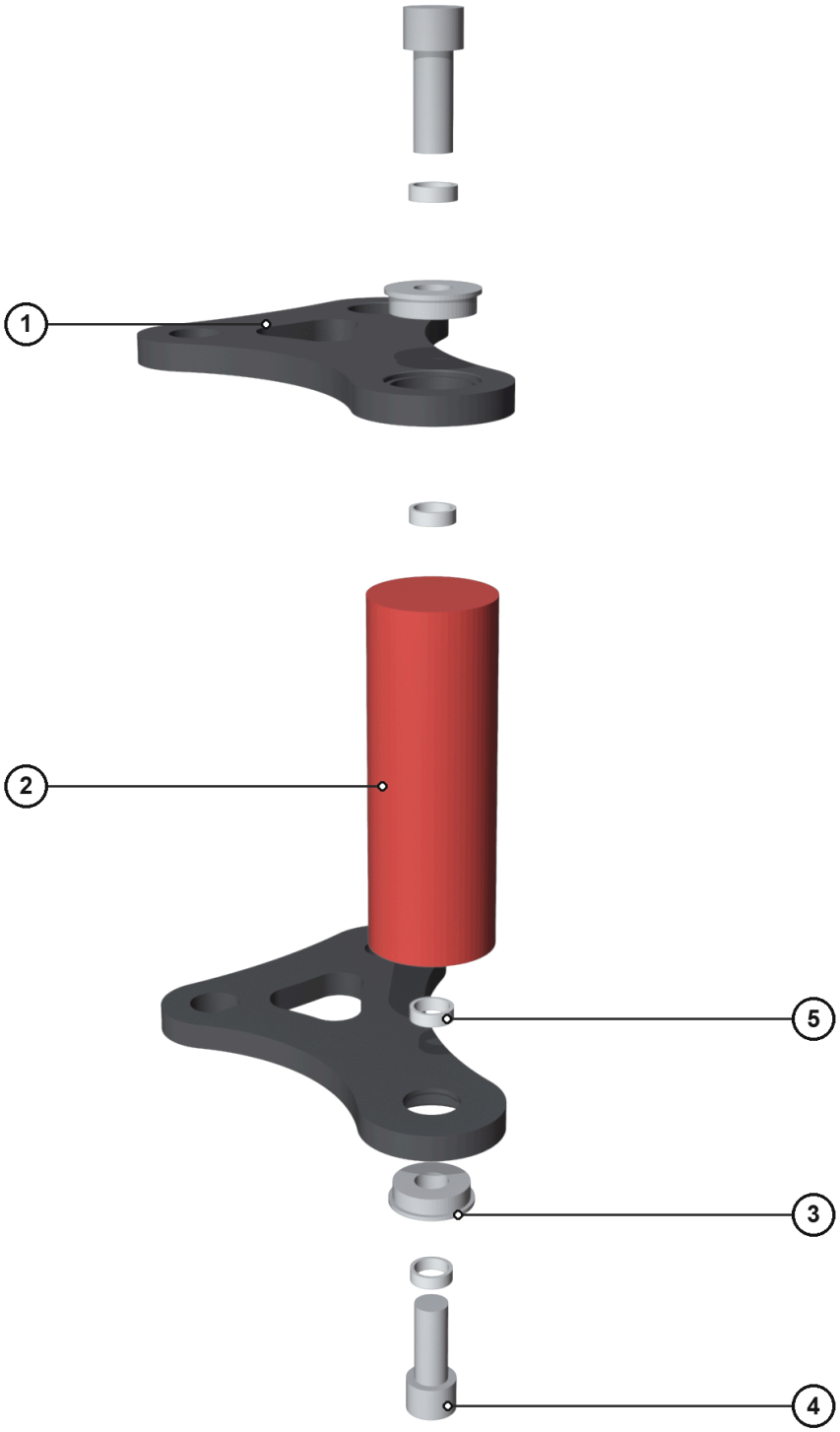
handle between the plates · pad behind · yoke outside, spinning



1	Side plate (printed) — PETG-CF, 10 mm	×2
2	Rogue H-5 handle 45 mm × 5" — 1/2"-13 both ends	×1
7	Carabiner yoke (printed) — clean the Ø12 pivot bores, press in both steel sleeves (21)	×1
8	Bronze sleeve ID 20 × OD 23 × 10, Ø30 flange — flange up, in the eye pocket	×1
11	Concave foam roller pad (trim to 132 mm) — slides over the dowel	×1
13	Button head 1/2"-13 × 2", black oxide — threads into the dowel, snug, NO Loctite	×2

2 · Handle assembly

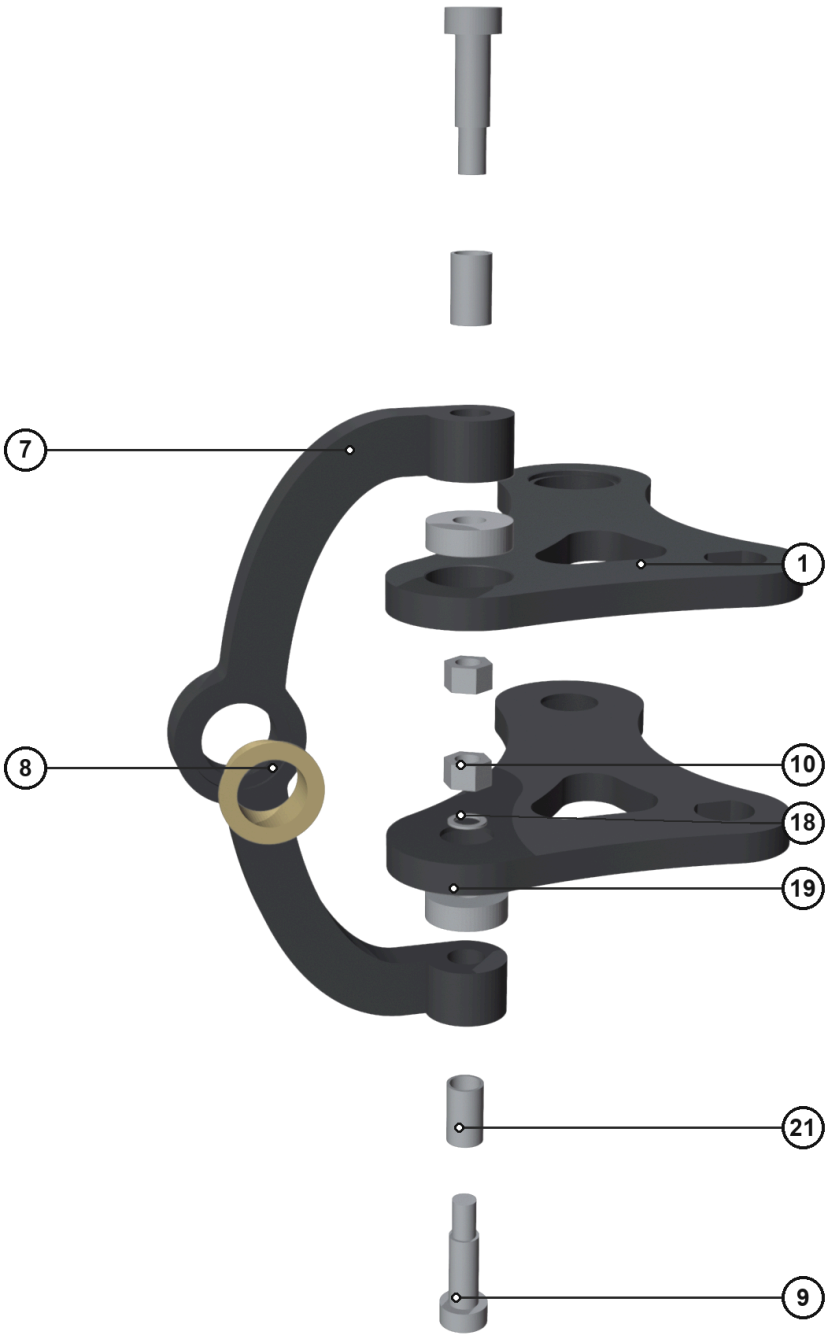
screw → Al spacer → bearing (in plate) → Al spacer → handle · both ends



1	Side plate (printed) — PETG-CF, 10 mm	×2
2	Rogue H-5 handle 45 mm × 5" — 1/2"-13 both ends	×1
3	Flanged bearing FR8-2RS — press fit from outside — no Loctite	×2
4	Button head 1/2"-13 × 1-1/4", black oxide — + blue Loctite, ≤ 15 ft-lb	×2
5	Al spacer 1/2 × 5/8 × 3/16" — chamfer the bearing-side edge 0.5 mm	×4

3 · Yoke assembly

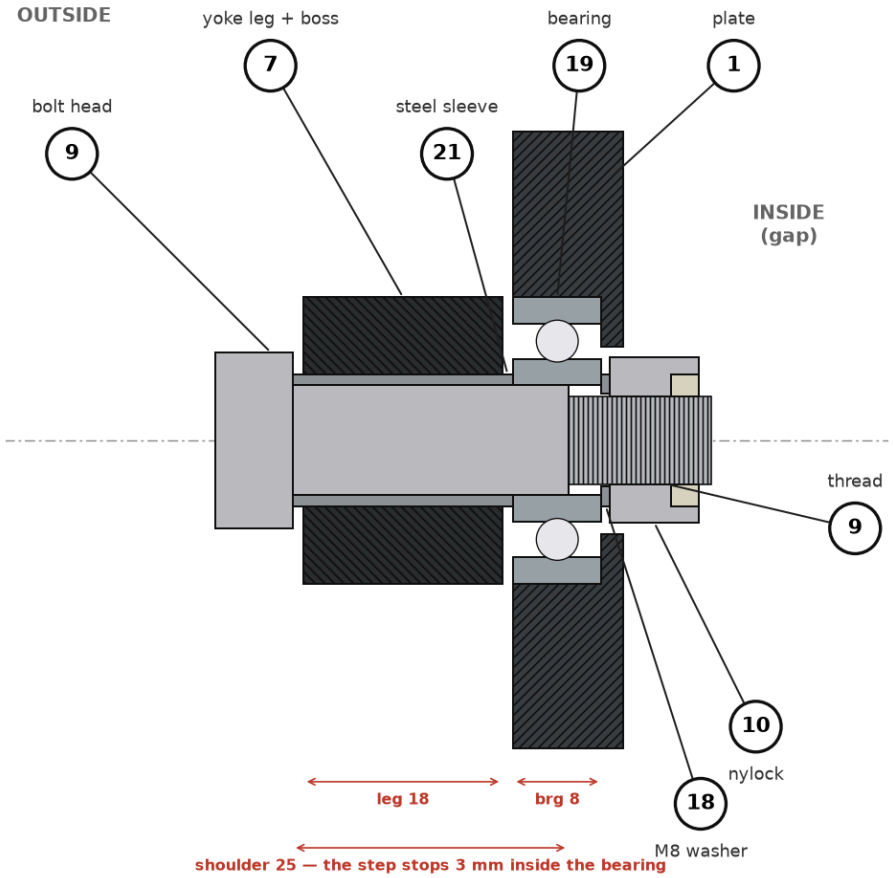
see 3b for the stack · bolts out, nylocks in · yoke straddles the plates



1	Side plate (printed) — PETG-CF, 10 mm	×2
7	Carabiner yoke (printed) — clean the Ø12 pivot bores, press in both steel sleeves (21)	×1
8	Bronze sleeve ID 20 × OD 23 × 10, Ø30 flange — flange up, in the eye pocket	×1
9	Shoulder bolt 10 × 25 mm, M8 — turns with the yoke	×2
10	M8 nylock (DIN 985) — ~4 N·m — see 3b	×2
18	M8 washer 8 × 12 × 0.8, stainless — under the nylock	×2
19	Bearing 6000-2RS 10 × 26 × 8 — press fit in each plate	×2
21	Steel sleeve SF-1 1020, 10 × 12 × 20 — 1 mm proud each face	×2

3b · Yoke pivot — section

cut through one pivot, to scale · the other side is the mirror image



What turns:

bolt + steel sleeve + bearing INNER ring + washer + nut: an all-metal clamp, spinning as one on the balls. The yoke is pressed on the sleeve and rides with it — the plastic is NOT squeezed.
Sleeve and washer are Ø12: inner ring only, clear of the seals.
The plate holds only the OUTER ring (seat Ø26 × 8, 2 mm ledge).

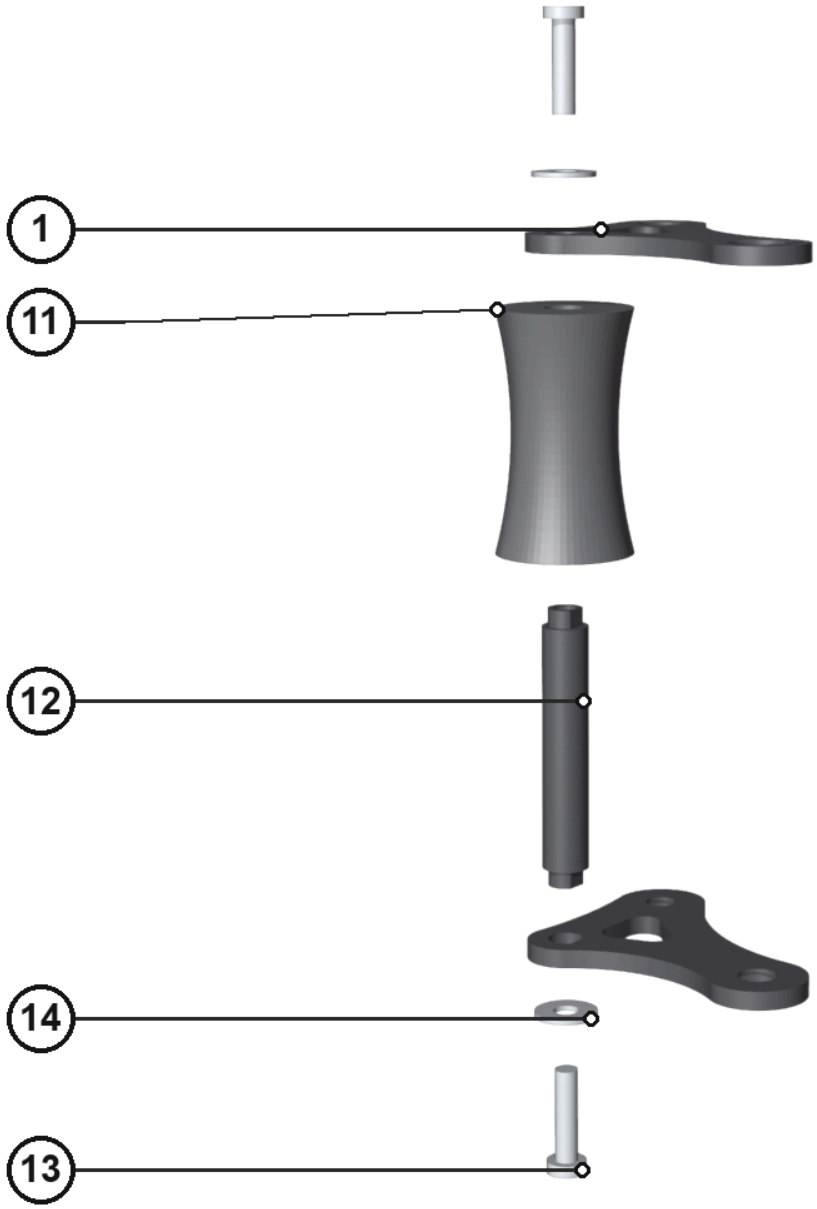
Assembly, each side:

1. Press bearing 19 into the plate's outside face, flush.
   Press fit — NO Loctite (it attacks PETG); epoxy if loose.
2. Clean the yoke bore to Ø12 (drill bit by hand), then press sleeve 21 through the boss, 1 mm proud each face.
3. Bolt 9 through the sleeve and the bearing; from INSIDE the plates add washer 18 and nylock 10.
4. Tighten to ~4 N·m (3 ft-lb) — firm on a short hex key.
5. Check the sleeve-to-sleeve span equals the plates' outside-to-outside (152.4 mm / 6.00"). If not, the yoke pre-loads the bearings: re-seat the sleeves 1 mm proud. Re-check after the first sessions.

Mirror image on the other plate.

4 · Pad assembly

pulled apart on the axle · a button head threads into each end of the dowel · D ends key it into the plates



1	Side plate (printed) — PETG-CF, 10 mm	×2
11	Concave foam roller pad (trim to 132 mm) — slides over the dowel	×1
12	Pad dowel (printed upright, 100 % infill) — 1/2-13 threads printed in	×1
13	Button head 1/2"-13 × 2", black oxide — threads into the dowel, snug, NO Loctite	×2
14	Washer 1/2" × 1-1/4" OD, black — one each side	×2